

methodologies, test environment setups, use of tools and simulators, and more is needed.

The IIoT incorporates machine learning and Big Data to harness sensor data. It holds great potential in manufacturing for quality control, supply chain traceability and overall supply chain efficiency. Since manufacturing companies have more complex and dynamic processes, availability of real-time data on critical parameters becomes more important.

Role of sensors in the IIoT

Sensors help in predictive maintenance, asset monitoring and data analytics to achieve sophisticated machine-to-machine (M2M) communication. Automation in processes like control, safety, operations management and asset optimisation increase the demand for sensors for measurement and analytics. Smoke, proximity, IR, image, temperature, piezo and optical sensors are the different types of IoT sensors required for an industrial automation system.

Networked data loggers combined with cloud solutions enable test and measurement specialists to provide a complete digitisation solution for bringing systems, machines, vehicles and more to the IoT. Long-term measurement and monitoring are required to obtain measurement data under real operating conditions. This makes it possible for engineers to have a completely-isolated personal testbed at their desk. Different tests can be performed in parallel by multiple engineers and results saved in a common database.

Test approaches for the IIoT

The primary focus of testing during industrial automation is ease of use, ease of learning or familiarising with the system, and satisfaction of the user with the entire experience. To ensure that scalability, performance and security is up to the mark, following types of tests are recommended:

Usability. The tracking device used should be portable. Usability in terms of processing and displaying data should be tested thoroughly.

It should be smart enough to push

not only notifications but also error messages, warnings and so on.

It should have the option to log all events, or store the same to a database, to provide clarity to end users.

Notifications and handling of the display should be done properly in the devices.

IoT security. All connected devices operate based on available data. There is always a chance that data can be accessed or read while transferring between devices. Hence, it is important that the user interface is password protected.

Before communication starts, the hub and devices should be properly authenticated. Sent data should be in encrypted form.

Connectivity. Success of an IoT system depends on how well the devices and hub are connected. Loss of connection can lead to inaccurate data.

Every authenticated device in the range should be able to connect to the hub when the connection is up and running. It should be able to send any required amount of data to the hub.

In case of a down connection, if the system is unavailable on the network, timely alerts should be sent.

Performance. It is important to test the monitoring utility used to display system usage, power usage, temperature and more. IoT devices in sleep mode draw nanoamps of current but may draw tens of milliamps while processing or communicating. To deliver power-efficient IoT devices to large utility customers on time and with real-world battery life performance, following conditions must be checked:

- When a device is located at cell edge and has intermittent connectiv-

ity, carrier aggregation, higher-order MIMO and other cellular-IoT capabilities' impact on battery lifespan must be calculated.

- Power status update should be sent to the network if the device goes low in power.

Reliability and scalability. Testing in a lab does not ensure that the system will work when exposed to real-time conditions. Hence, a test environment must involve a simulation of sensors, for reliability and scalability, by leveraging virtualisation tools and technologies.

Other than testing in real-life scenarios, there are a few common test cases that must be considered while testing IoT devices and the network.

Regulatory. It is a good practice to get regulatory requirements approved at the start of the development cycle, to make sure that the products are certified.

Upgrade. When an upgrade is performed, be it for the system or for any involved item, thorough regression testing should be carried out to overcome upgrade-related issues.

Signal and power integrity. The IoT is successful if the integration test plan is accurate and robust enough to catch flaws in the system. Devices need carefully-controlled signal and power integrity.

Compatibility. There is a large number of devices available for particular applications of sensors, protocols and platforms. As a result, a test matrix is formed to achieve the right compatibility testing.

Communication protocol and device interoperability. This assesses the ability to interoperate protocols and devices across different standards and specifications.

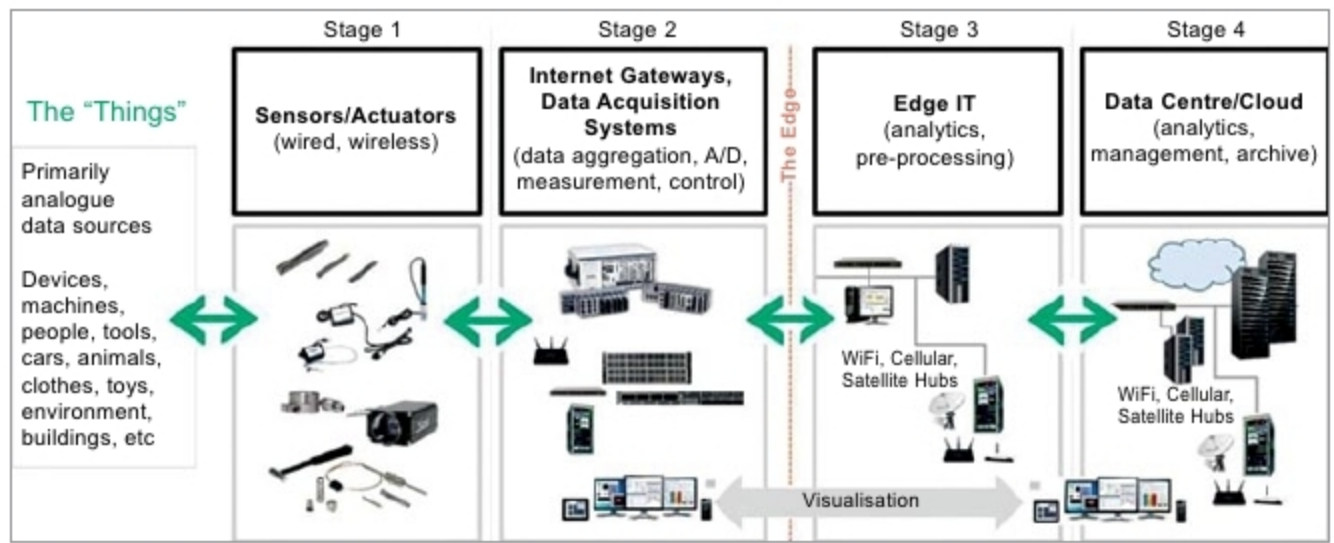


Fig. 1: A four-stage IIoT solution (Credit: www.sciencedirect.com)